We're proudly pushing forward the PostScript Pareto Frontier.

Every TONER model runs inside a print job. We plotted everyone else for scale.

FIGURE 1 THE FRONTIER (PARODY)

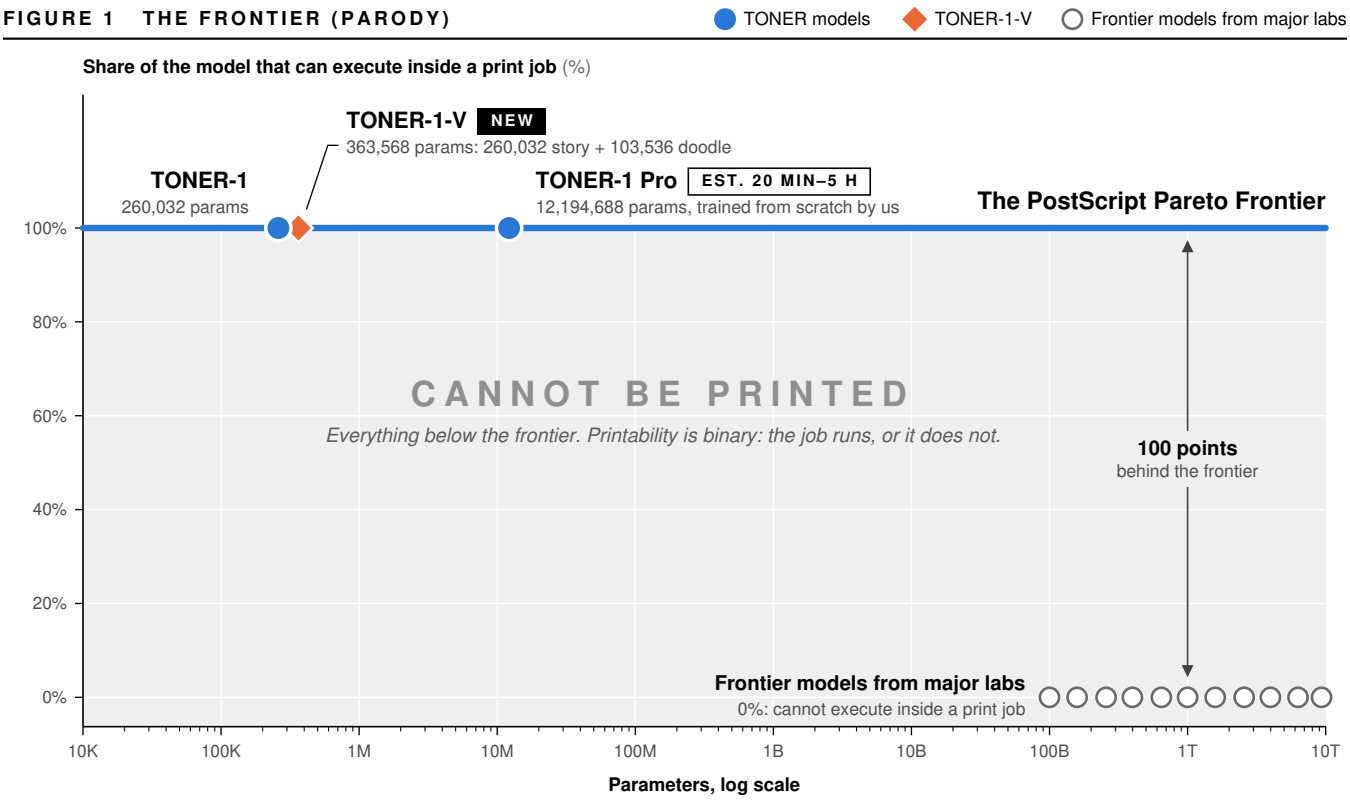


Figure 1. Every TONER model sits on the frontier at 100%: weights, vocabulary and forward pass all execute inside the print job. Frontier models from major labs, at roughly 400,000 to 40 million times the parameters of TONER-1, sit at 0%. At every parameter count where we have a model, the nearest competitor trails by 100 points.

BY THE NUMBERS

MODEL	PARAMETERS	PRINTABLE	STATUS
TONER-1	260,032	100%	Available
TONER-1-V	363,568	100%	Available
TONER-1 Pro	12.2M	100%	Available
Frontier models from major labs	100B to 10T	0%	N/A

WHY THIS MATTERS

- **Efficiency.** TONER-1 reaches the frontier with 260,032 parameters. Others use hundreds of billions to miss it.
- **Deployment.** One file, sent raw to hardware already in your copy room. Zero host FLOPs.
- **Multimodality.** TONER-1-V adds a doodle to the same page and job for 103,536 more parameters. On a log axis, that is free.

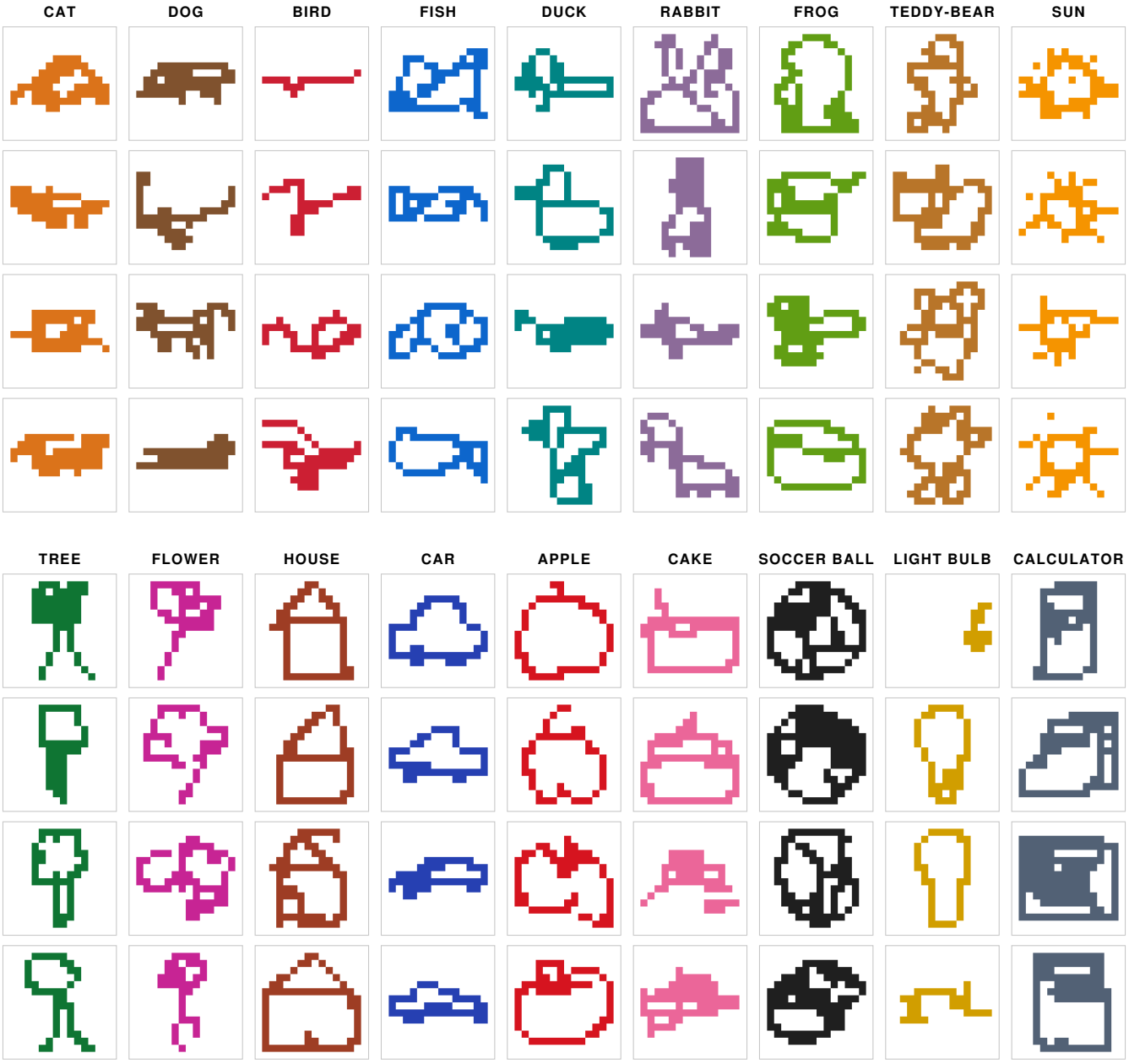
Parody chart. Not a real evaluation. Competitor parameter counts are illustrative. Frontier models from major labs were not run on this chart and are not named; 0% means only that they cannot execute inside a print job. TONER-1-V is plotted at 363,568 parameters: the story model plus the 103,536-parameter doodle model.

TONER-1 is a working print job and a parody of frontier-model launches. This handout is a PostScript program, too.
Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.

TONER-1-V sample sheet

Eighteen classes, four doodles each, every pixel drawn inside this one print job.

16 × 16 one-bit pixels each: after BOS and a class token, TONER-1-V samples 64 patch tokens (2 × 2 pixels, 16 possible values) at temperature 0.9, one forward pass per token, with a 103,536-parameter int8 transformer. Seeds 1–72, top to bottom within each class; unedited, nothing cherry-picked.



DOODLES	PATCH TOKENS	SECONDS	SEC / DOODLE	TOKENS / SEC	TEMPERATURE
72	4,608	22.0	0.31	210	0.9

INTERPRETER GPL Ghostscript · revision 10071 · PostScript version 3010

Seconds are interpreter usertime spent generating.

TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little boy named Tom. He loved to race with his friends. One day, Tom and Dog went to the park to play. They saw a big box with a coat. Tom wanted to print a box of copies first. Another copy goes, and a birthday color heard the creek.

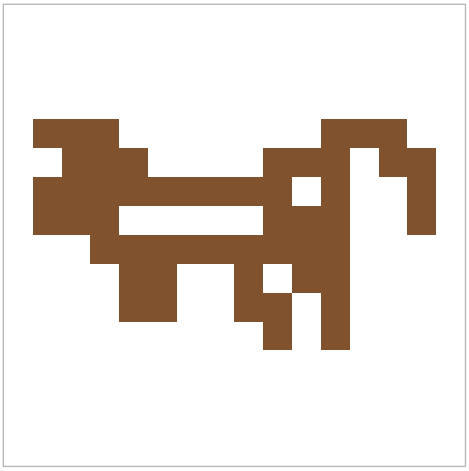
At night, Tom went to the park. “I need help!” said Tom. “I did it!” said Tom. Tom quickly took the paper and put it in the chair. Tom looked at the crum…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . . . . . .
. . . . . . . .
4 f 2 . 1 7 d 2
5 f c c d 7 . a
. 4 f c e f . .
. . c . d 5 . .
. . . . . . . .
. . . . . . . .
```



TONER-1-V’s drawing of a dog: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: “dog”, word 23 of 89. Class token 19 (dog).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K printer-tuned				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	81.0	7	0.8	64	256	7	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).

TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

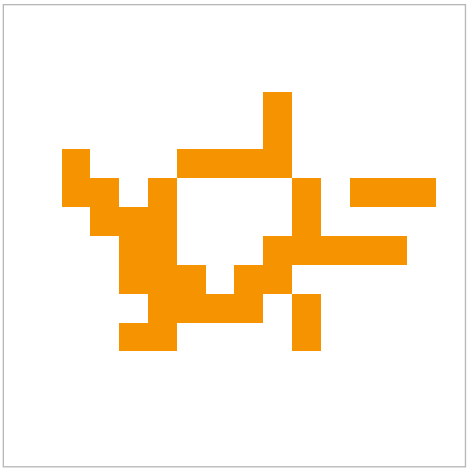
One day, a little girl named Leo wanted to print a picture of a sun. She pressed the button and put the paper into the printer. The printer hatt the paper came out wrong. Inky watched the paper out. Leo sat quietly with it. “I am here,” said Leo. “I can’t find it.” He picked up the papers. He put one piece of paper. The printer hummed. It printed a piece of paper. N…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . . . . . .
. . . . 1 . . .
. 2 . 3 7 . . .
. d 7 . . a c 8
. . f 2 7 c c .
. . 7 c 8 a . .
. . . . . . . .
. . . . . . . .
```



TONER-1-V’s drawing of a sun: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: “sun”, word 15 of 74. Class token 26 (sun).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K printer-tuned				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	80.6	4	0.8	64	256	4	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).

TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

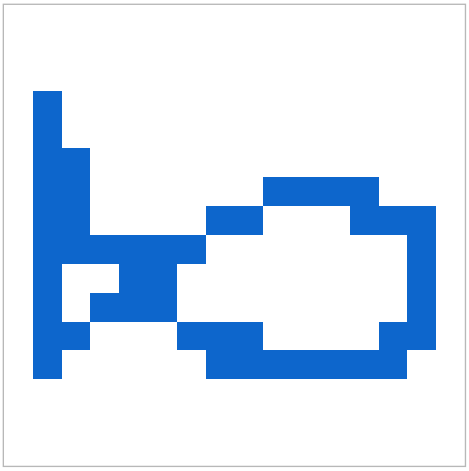
Mia and a big fish went to a sea. Lily wanted to see the sea. She pushed the button with it. But the printer mixed the seed inside. “What easy?” her mom asked. The seed was out quiet and did not feel neat. Liza picked it up and held it up the seed. “This is a fish!” he said. Then Mia walked into the room. She saw the paper and crossed the button. She pulled it out of the printer. It was not looking. M…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . . . . . .
1 . . . . . . .
5 2 . . . . . .
5 a . 1 6 c b 2
5 c f 8 . . . a
5 6 c 3 2 . 1 a
4 . . 4 c c c .
. . . . . . . .
```



TONER-1-V’s drawing of a fish: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: “fish”, word 5 of 85. Class token 21 (fish).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K printer-tuned				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	80.6	6	0.8	64	256	6	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).

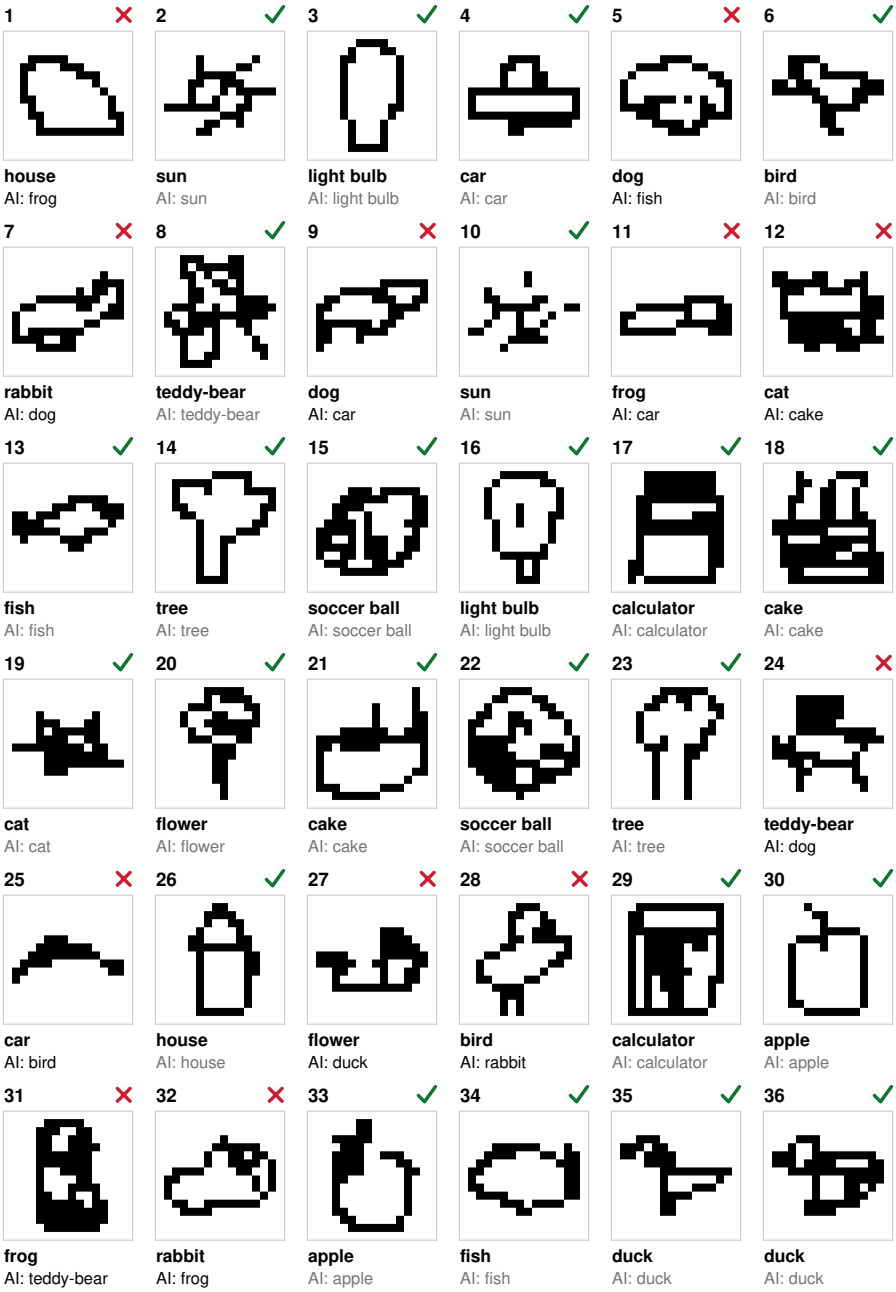
TONER-1-V: blind identification test

24 of 36 named correctly. Chance would get 2. Dogs, rabbits and frogs went 0 for 6.

The blind grid's 36 doodles, regenerated by this print job from the same seeds and drawn in black as the evaluator saw them, with each true class and the evaluator's guess. **Identified blind by an AI model that was not told the labels; not a human study.**

FIGURE 1 THE 36 DOODLES, WITH ANSWERS

✓ identified ✗ missed



RESULTS

24 of 36

doodles named correctly: 66.7%

CHANCE	1 in 18 = 5.6% (2 of 36)
95% INTERVAL	50%–80% (Wilson)
OBJECTS (10 CLASSES)	17 of 20 (85%)
ANIMALS (8 CLASSES)	7 of 16 (44%)
AI: RECOGNIZABLE	18 of 18 right
AI: NOT RECOGNIZABLE	6 of 18 right

PER CLASS

■ identified □ missed

fish	■■	
duck	■■	
sun	■■	
tree	■■	
apple	■■	
cake	■■	
soccer ball	■■	
light bulb	■■	
calculator	■■	
cat	■□	as cake
bird	■□	as rabbit
teddy-bear	■□	as dog
flower	■□	as duck
house	■□	as frog
car	■□	as bird
dog	□□	as fish, car
rabbit	□□	as dog, frog
frog	□□	as car, teddy-bear

THIS PAGE

Every doodle here was regenerated inside this print job from the blind grid's seeds: 2,304 patch tokens in 10.8 s, 213 tok/s. **36 of 36 match the blind grid token for token.** Interpreter: GPL Ghostscript · revision 10071 · PostScript version 3010.

Identified blind by an AI model that was not told the labels; not a human study. The evaluator reports reading only the unlabeled blind grid page, which lists the 18 classes and says there are two of each; it did not force the pairs (cake, car, duck and fish got three guesses each). Scoring is an exact match against the answer key; "AI: recognizable" is the evaluator's own flag. 36 doodles is a small sample, and with two per class the per-class results are anecdotes. For scale, the CNN judge used in training scores 77.8% on 288 Ghostscript doodles at the same temperature and 81.4% on real 16 × 16 drawings.